5.1 APP Control

1. APP Installation

iOS users: search “Wonderbot” in App Store and download it.

Android users: find the “Wonderbot” pack in the same directory of this lesson or refer to the appendix to install it.

2. Device Connection

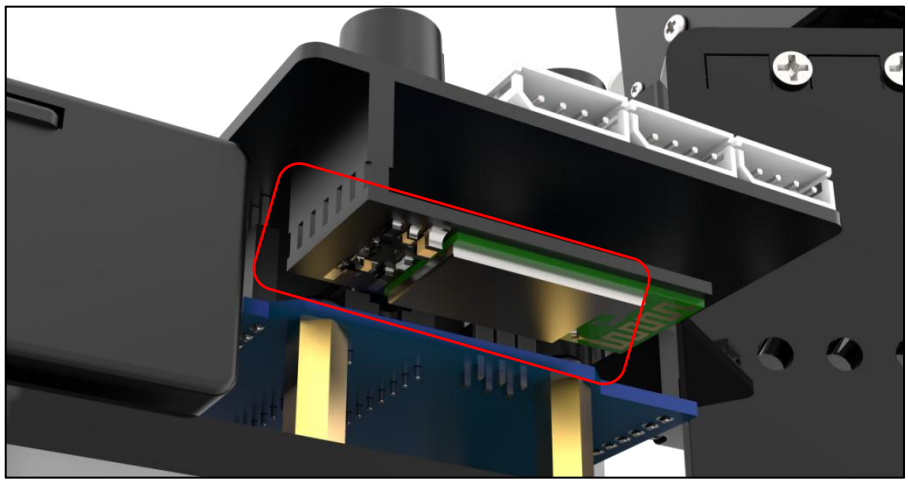
Note

1) Please enable Bluetooth and location services in the phone settings before opening APP.

2) Please use the Bluetooth button in the APP to pair and connect with the device. Do not pair with the key in the phone settings.

3) Take Android APP as example in this lesson.

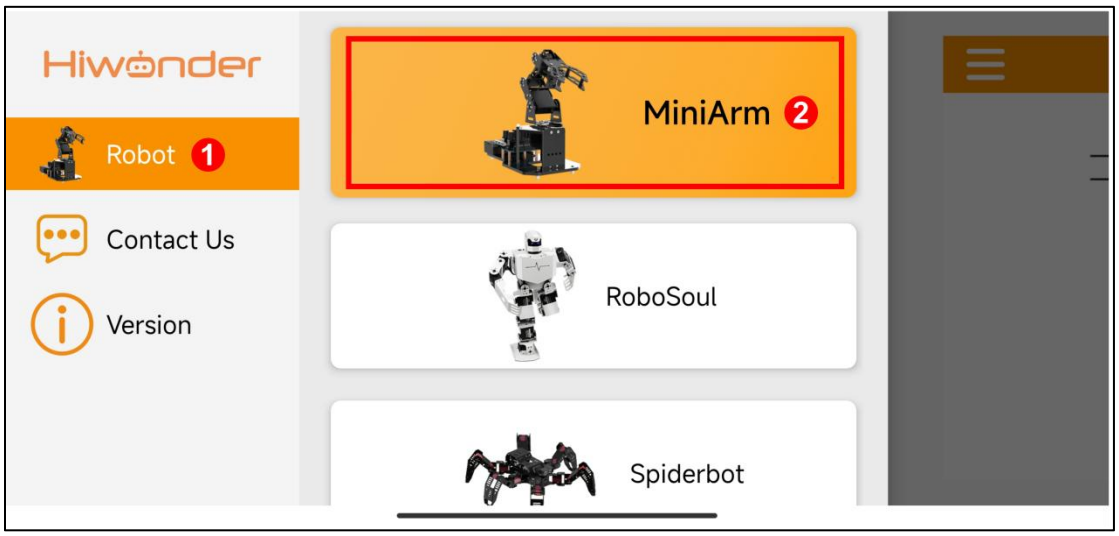
Step 1: Make sure the Bluetooth module is installed correctly on the miniArm 6 channel knob expansion board by aligning pins one by one, as shown below:



Step 2: Turn on the power switches of the battery case and the expansion

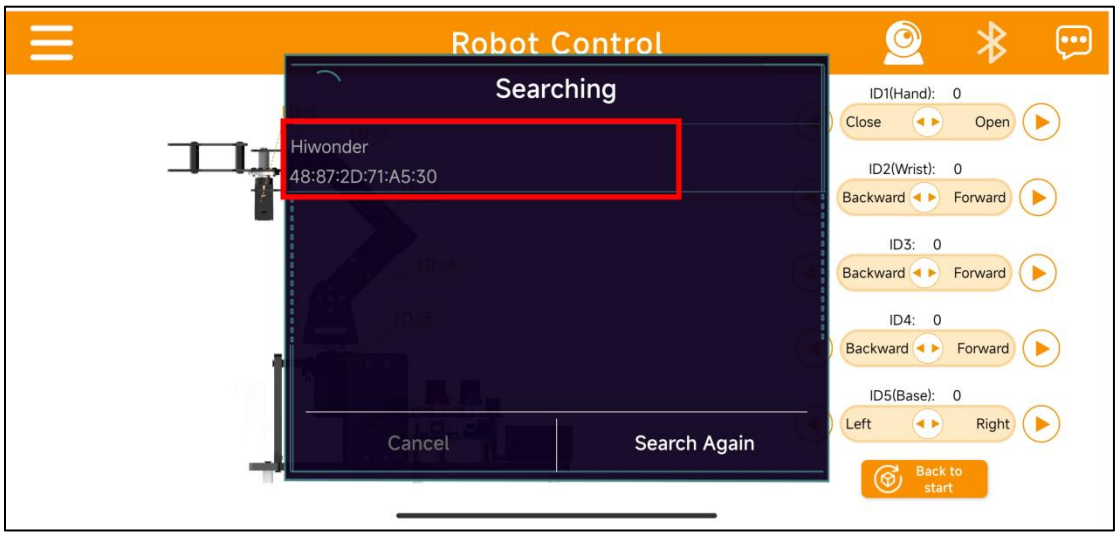
board. Press the K1 and K2 buttons on the expansion board at the same time, and release them until the RGB light turns to green. Then it can exit the neutral position mode. (If it does not exit, you can not control by the knob or app.)

Step 3: Open the APP“Wonderbot”, and tap on the icon at the top left corner to select “MiniArm”.

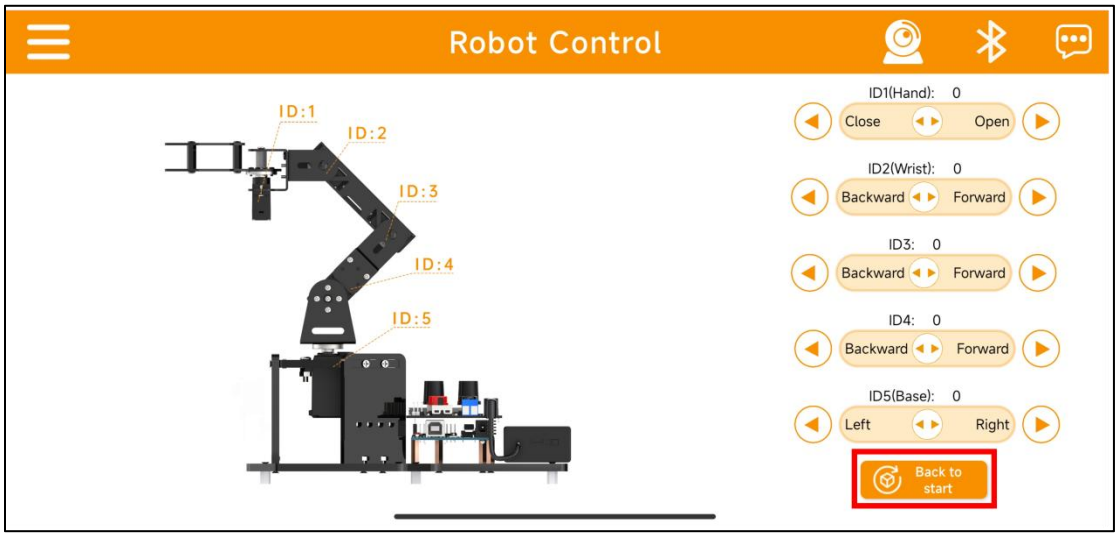


Step 4: In game interface, tap on at the top right corner, and find "Hiwonder" in Bluetooth list. Click on the link.

! If the "Hiwonder" does not appear immediately after clicking on the Bluetooth icon, you can click "Retry" to search it again.



Step 5: After successfully connecting, the Bluetooth icon at the top right corner will switch to a steady mode. Click "Back to start" to return the robotic arm to the initial position.

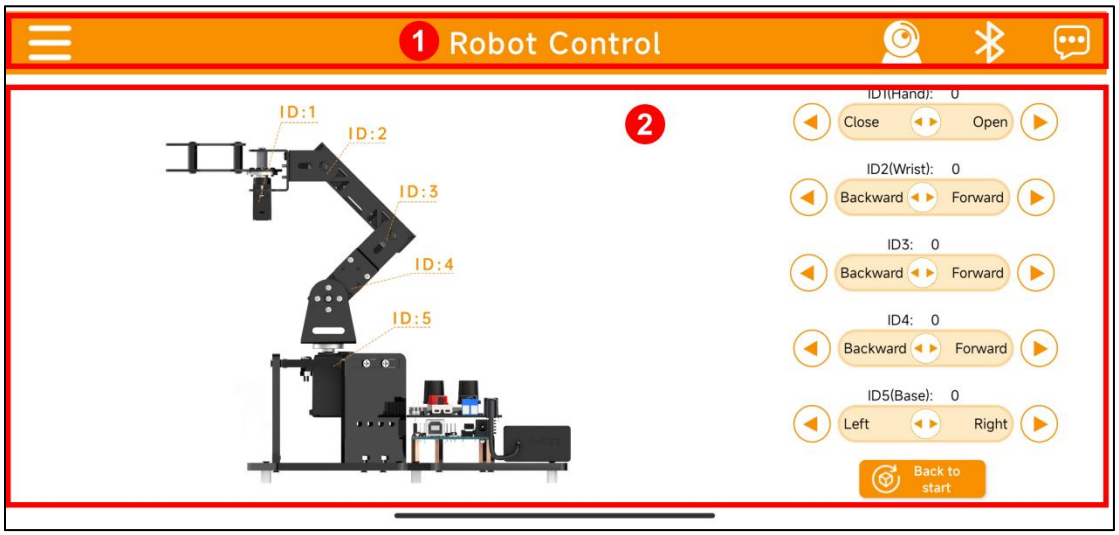


3. Control Instruction

3.1 Slider Control

The app has a slider control function and live camera feed function. You can control rotation angle of each servo of the robotic arm by sliding the slider.

The app interface consists of two parts as shown below:



①：Menu Bar

	Return to the main interface to select robot type
	Open the live camera feed function
	Bluetooth connection
	More information

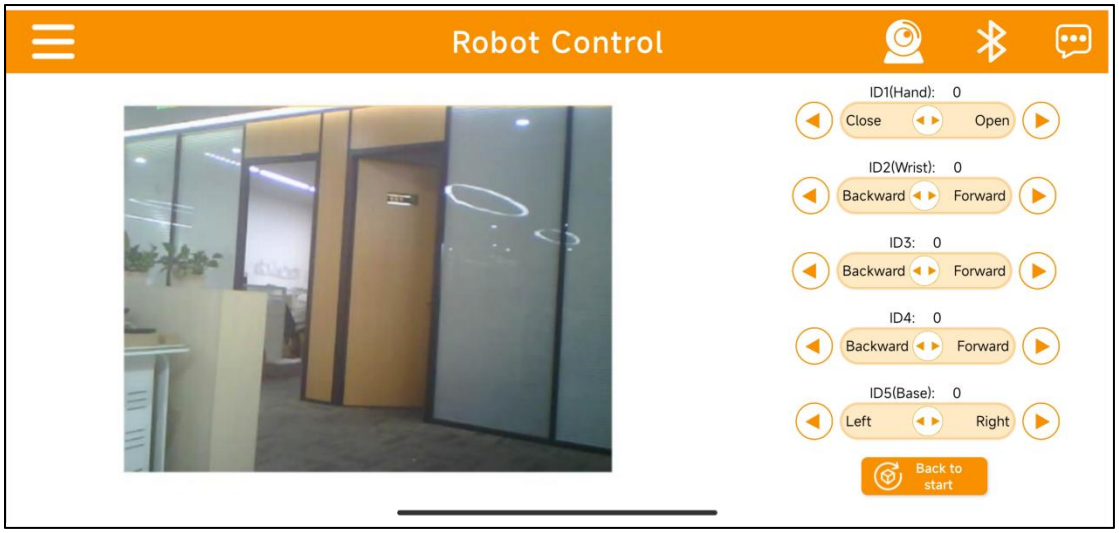
②：Control Area

	Servo No.1 controls the robotic claw to open and close, with the angle range of 0 to 180 degrees
	Servo No.2~5 control the rotation of robotic arm, with the angle range of 0 to 180 degrees
	Control all the servos to return to the initial position

3.2 Live Camera Feed Function

Note: To achieve the live camera feed function of the miniArm, please flash the firmware into ESP32S3-Cam based on the relevant tutorial.

1.Click icon on the menu to open live camera feed mode. The interface of this mode is as pictured:



Now you can see the real-time image transmitted by ESP32S3-Cam on the app.